Danielis Maizelis

Embedded System / Electronics Engineer

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Summary

Embedded Software Engineer specializing in bare-metal C and automotive communication protocols (CAN/LIN). Delivered firmware for ECU diagnostics and QUAD CAN/LIN testing at Elinta Motors, and independently designed, trained (PPO/deep RL), and flight-tested an autonomous drone system end-to-end. Comfortable moving across the full stack — from PCB and low-level firmware to Python/ML — and picking up new hardware or protocols quickly.

Experience

Elinta Motors — *Junior Embedded Software Engineer*
Kaunas, Lithuania · February 2023 – May 2023

- Took ownership of the QUAD CAN/LIN tester project for an electric drivetrain, delivering a working diagnostic tool used for hardware validation
- Documented troubleshooting procedures for recurring hardware and software issues, now used by the team for ongoing debugging

Elinta Motors — *Embedded Software Engineer Intern*
Kaunas, Lithuania · January 2023 – February 2023

- Soldered SMD components onto PCBs for electric drivetrain testers
- Wrote CAN and LIN firmware for STM32-based microcontrollers.

Education

Kristianstad University, Kristianstad, Sweden — *Bachelor in Computer Science, Specialisation in IOT (*September 2023 – June 2026)

- Wrote a paper on PPO reinforcement learning using autonomous drones. Trained the 500mm drone using Deep Reinforcement Learning with Unreal Engine, then assembled and flight-tested it in real life.

Business Academy Aarhus, Aarhus, Denmark — *AP Degree in Embedded Systems (*September 2021 – June 2023)

Skills

- **Firmware & Embedded:** C (Bare-Metal), Assembly, STM32, AVR/ATmega, UART, CAN, LIN, I2C, SPI, Bluetooth, Zigbee.
- **Software & ML:** Python, Java, Kotlin, Machine Learning, Deep Reinforcement Learning,
- **Microcontroller boards:** Jetson Orin Nano, Raspberry Pi, Arduino, ESP32.
- **Hardware:** PCB Design, Soldering, 3D Printing, Hardware Debugging.
- **Tools & OS:** Linux, VIM
- **Certifications:** A1/A3 Drone License, B1 Driver's License
- **Languages:** Lithuanian, English, Russian, Hebrew
- **Cloud:** Microsoft Azure (Blob Storage)

Projects:

Done Projects

RL Forest Drone — *Reinforcement Learning · 2026*

- Trained a PPO policy (Stable-Baselines3, custom CNN) to fly a drone through procedurally-generated forest using only monocular RGB — no LIDAR, no stereo cameras
- Built the simulation in Unreal Engine/AirSim, then transferred sim-to-real onto a Pixhawk 6C/PX4 airframe with a Jetson for onboard inference in a real-life forest. Used Depth Anything V2 vision model for interpreting objects.
- GitHub: github.com/DanielisMaizelis/redwood_airsim

Daniel Tube — Python · *Terminal Web App · 2026*

- Built a terminal-based YouTube client that downloads and auto-translates video subtitles using the Whisper API
- Stack: Python, yt-dlp, Whisper, FFmpeg
- GitHub: github.com/DanielisMaizelis/danieltube

Mum's List — *Embedded Systems · ESP32 · 2023*

- Built an IOT fridge inventory tracker using barcode scanning and weight sensors to detect remaining product quantity

Robocar — *Embedded Systems · AVR Bare Metal · 2025*

- Programmed an autonomous line-following robocar in bare-metal C on an ATmega328P, which connects via Bluetooth.

Axis Garage Parking App — *Kotlin Mobile App· 2023*

- Built a Kotlin app integrating AI parking-spot cameras via MQTT, with Google OAuth and role-based access

GTA Radio — *PCB For a car · 2023*

- Custom STM32-based car radio PCB with 12 to 5V SMPS, Micro SD playback, FM tuning, making sure components do not interfere with one another.

Ongoing Projects

- **GTA Radio PCB Firmware** — Custom STM32-based car radio Firmware with SMPS, Micro SD playback, FM tuning
- **Class AB Amplifier Circuit** — PCB design exercise
- **SPI Poker** — Electronic poker game reading RFID cards over SPI, that evaluates each of the players' chances.